Sr.No. 8374

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:202 Course Section:1	Seat No: 2022600008

L_61_466_986_1_1505_48190

Name of the candidate. Vaishnavi Prabha Ramesh Chettiar

Candidate's UID number: 2 0 2 2 6 0 0 0 0 8

Examination class room number: 2 0 1

Invigilator's signature with date: 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BE BTECH : MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE - AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 8-10-25

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
3) Do not write anything in the column provided for the marks to be assigned by the examiners.
4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
12) The answer-books will be scrutinized before they are sent to examiners.
13) All answer-books supplied shall be returned whether written or blank.
14) Nothing shall be written on the question-paper.
15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	
	Q1a	$J(w) = \frac{1}{2} \|\|Xw - y\|\|^2$

gradient update rule:

$W_{new} = \cancel{W_{ne}}\ W_{old} - \eta \Delta w$.

where Δw is partial derivation of Loss function w.r.t w (weight).

$\therefore \Delta w = \frac{dL}{dw}$.

Now $\frac{dL}{dw} = \frac{dL}{da} \cdot \frac{da}{dz} \cdot \frac{dz}{dw}$.

$L = \frac{1}{2} \Sigma |\hat{y} - y|^2$ — MSE.

$\hat{y} = a$ (activation f^n) $= \frac{1}{1+e^{-z}}$.

$z = wx + b$.

Now $\frac{dL}{da} = \frac{d}{da}\left(\frac{1}{2} \Sigma |a - y|^2\right)$.

$= \Sigma a - y$

$= (A - Y)$.

$\frac{da}{dz} = \frac{d}{dz}\left(\frac{1}{1+e^{-z}}\right) = \frac{+\cancel{1}e^{-z}}{(1+e^{-z})^2} = \frac{1+e^{-z}-1}{(1+e^{-z})^2}$.

$= \frac{1}{1+e^{-z}} - \frac{1}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}}\left(1 - \frac{1}{1+e^{-z}}\right)$.

$= a(1-a) = A(1-A)$.

$\frac{dz}{dw} = wx + b = x = X$.

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$$\therefore \frac{dL}{dw} = (A-Y)(A(1-A))(X).$$

$$\therefore W_{new} = W_{old} - \eta\left[(A-Y)(A(1-A))(X)\right]$$

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Space For Marks	Question No.	
	Q3b	

~~To~~ The eqn^s of GRU can be stated as :

$$h_t = u_t \cdot \tilde{c}_t + (1-u_t) \cdot h_{t-1}.$$

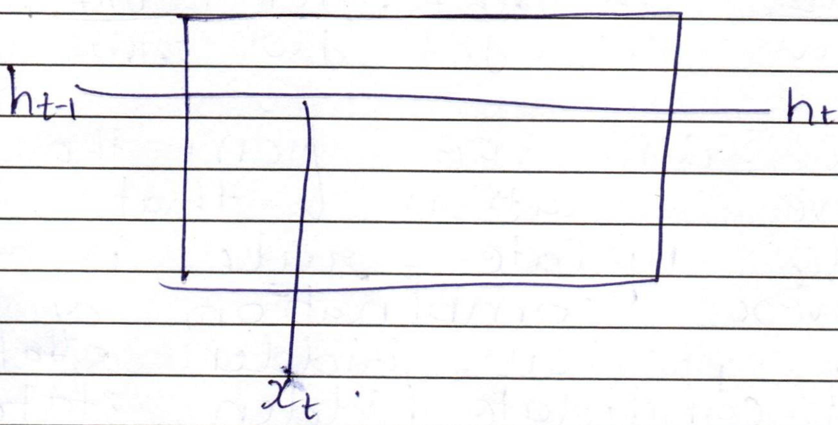
where u_t = update gate
$\tilde{c}_t$ = candidate hidden state.
r_t = reset gate.

now $u_t = \sigma(W_u(h_{t-1}, x_t) + b_u)$.

$r_t = \sigma(W_r(h_{t-1}, x_t) + b_r)$.

$\tilde{c}_t = \tanh(W(r_t \cdot h_{t-1}, x_t) + b_c)$.

GRU cell looks like :



Now the ~~of~~ to get the update the loss function depends on h_t.

$\therefore \frac{dL}{dh_t}$.

to get upto weight we have to apply chain rule :

as we can see the the h_t eqn depends on $\tilde{c}_t, h_{t-1}, u^t$
but u_t again depends on h_{t-1}.
$\therefore$ h_t depends solely on $\tilde{c}$ & h_{t-1}

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Therefore

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial h^t} \cdot \frac{\partial h_t}{\partial w}$$

now $\cancel{\frac{\partial h_t}{\partial w} = \frac{\partial \tilde{c}}{\partial w} + \frac{\partial h_{t-1}}{\partial w}}$

$\therefore \cancel{\frac{\partial L}{\partial w} = \frac{\partial L}{\partial h^t} \cdot \left(\frac{\partial \tilde{c}}{\partial w} + \frac{\partial h_{t-1}}{\partial w} \right)}$

now $\frac{\partial h_t}{\partial w} = \frac{\partial h_t}{\partial \tilde{c}_t} \cdot \frac{\partial \tilde{c}_t}{\partial w} + \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial w}$.

$$\therefore \frac{\partial L}{\partial w} = \frac{\partial L}{\partial h^t} \left(\frac{\partial h_t}{\partial \tilde{c}_t} \cdot \frac{\partial \tilde{c}_t}{\partial w} + \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial w} \right).$$

∴ we can see from the above equation that the update rule is a convex combination of the previous hidden state and candidate hidden state.

- GRUs just has ~~h~~ hidden state where as LSTM has Memory cell state as well as hidden cell state
- GRU is just ~~a~~ simplified version of LSTM

LSTM eqⁿ are given as

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t$$

where f_t = forget gate
i_t = input gate
c_t = Memory cell state

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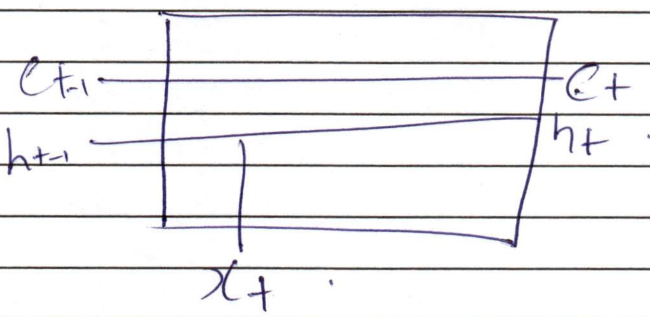
O_t = output gate
h_t = higgden state
$\tilde{C}_t$ = hidden candidate state

$$f_t = \sigma(W_f(h_{t-1}, x_t) + b_f)$$
$$i_t = \sigma(W_i(h_{t-1}, x_t) + b_i)$$
$$o_t = \sigma(W_o(h_{t-1}, x_t) + b_o)$$
$$\tilde{C} = \tanh(W_c(h_{t-1}, x_t) + b_c)$$
$$h_t = o_t \cdot \tanh(C_t)$$



We can see that GRU just has 4 eq^n and LSTM 6 eq^n hence we can say that GRU has fewer parameters than LSTM.

- Since C_t has been removed from GRU it was in LSTM to get long term memory dependency now GRU incorporate that in a single hidden state using update gate ~~state~~ and reset gate

- Here upate gate is used ~~wet~~ what part of memory to remember and what part of input we need to store
- ~~This step~~ And reset gate is used to forget all unnecessary past memory

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		- The use of these 2 state helps generalization in real world application

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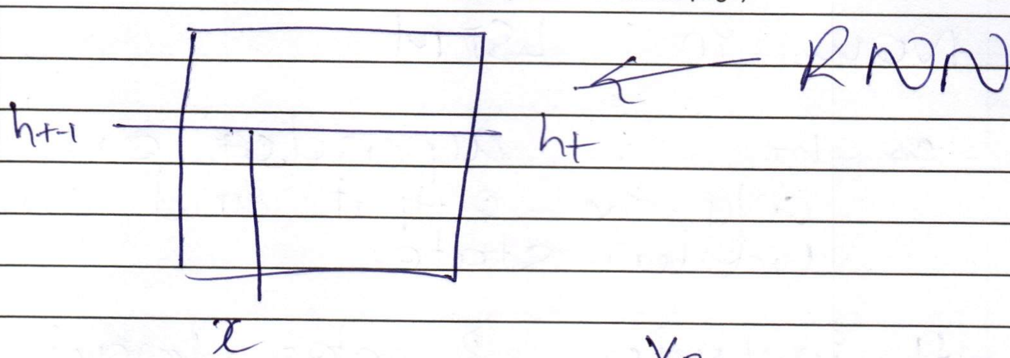
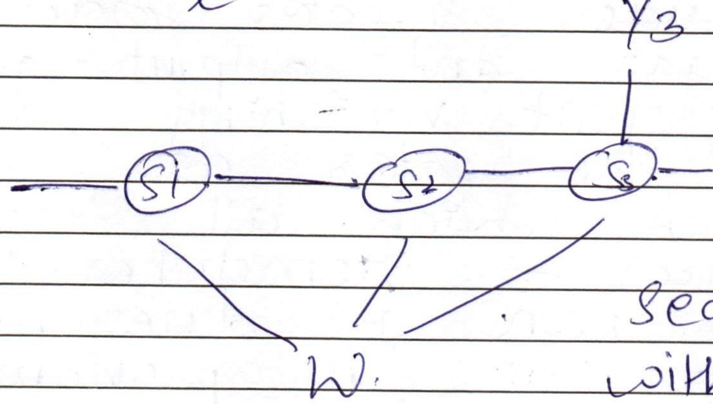
	Q.3a	





Sequential RNN with 3 cell.

$L = |Y_3 - Y|^2$ To get $\frac{dL}{dW}$.

Loss function depends ~~of~~ on Y_3. so partially differentiating w.r.t Y_3

$\frac{dL}{dY_3}$

Y_3 is depends on all S_1, S_2, S_3

$\therefore \cancel{\frac{dL}{dY_3}}$

$$\frac{dL}{dW} = \frac{dL}{dY_3} \cdot \frac{dY_3}{dW}$$

$$\frac{dL}{dW} = \frac{dL}{dY_3} \sum_{i=0}^{3} \cancel{\frac{dL}{dY}} \frac{dY_3}{dS_i} \frac{dS_i}{dW}$$

To generalize for n RNN cell we can say

$$\frac{dL}{dW} = \frac{dL}{dY_3} \sum_{i=0}^{n} \frac{dY_3}{dS_i} \frac{dS_i}{dW}$$

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Now in LSTM

ht is depends on only the output and candidate hidden state

- It includes 3 gates forget gate, input gate and output gate to prevent vanishing gradient
- since the gates are used to check which all state or sequence to remember unlike RNN it does not depends on all previous cell
- There by preventing vanishing gradient upto some extent

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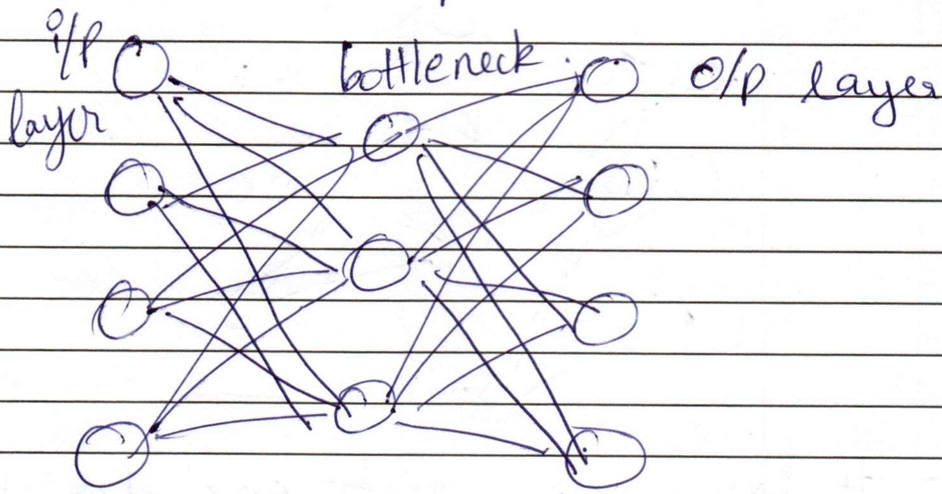
	Q2b	

Undercomplete auto encoder



- It is an auto encoder whose bottleneck or hidden layers dimension is less than that of input layer
- This is used to ~~comps~~ reconstruct input by compressing and extracting only essential factors.

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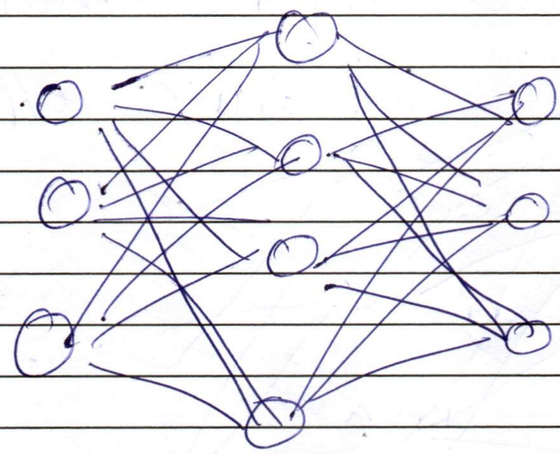
Q2b

Over complete Autoencoder



- It is an auto encoder whose bottleneck layers ~~or input layer~~ dimension is more than input layer.
- So the o/p layer copies i/p layer without removing any noise thereby calling it as an identity function
- ∴ there it is a need to introduce regularization inorder to denoise it

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	Q1b	- Overfitting is the tendency of model to learn all the patterns along with the outliers thereby memorizing the entire data rather than detecting any pattern
		- So when overfitting occurs the variance is high and bias is low
		- So to reduce the model capacity we introduce regularization, dropout layer inorder to reduce the variance to lower the high model capacity.

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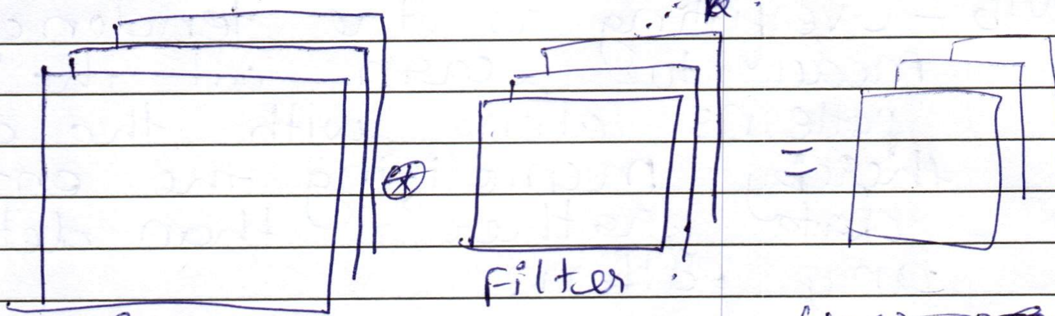
	Q2a	



Filter

~~(H-F+1)~~ ~~×K~~

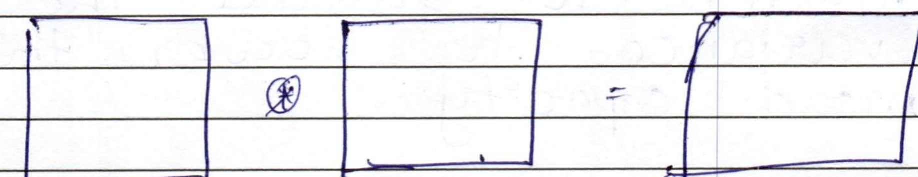
i/p

$H \times W \times C$

$F \times F \times \cancel{C}$. ~~A+P~~

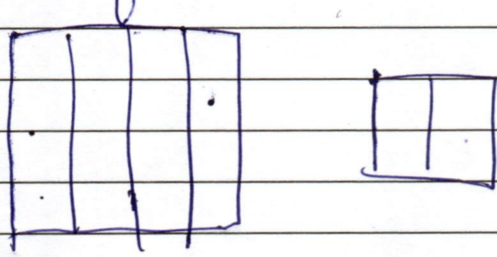
K Filters $(H-F+1) \times (W-F+1) \times K$

For a 2D convolution



$A \times B$ $\quad f \times f \quad$ $(A-f+1) \times (B-f+1)$

let $A = 4$ $\quad B = 4$.
$f = 2$.



the number of multiplication is 9. so to generalize it we can say

$(A \times f + 1)$ multiplication.

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